

ANTHROPOLOGY

A mid-Holocene age for Monte Verde challenges the timeline of human colonization of South America

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Our understanding of the timing of the human colonization of South America has been anchored by the Monte Verde II site in Chile, reported to date to ~14,500 years before the present (B.P.) and regarded as one of the most secure pre-Clovis archaeological sites. We report the first independent investigation of Monte Verde in the nearly 50 years since initial excavations. We argue that radiocarbon and luminescence dates from alluvial exposures, in combination with the identification of a tephra dated to 11,000 years B.P. stratigraphically underlying the archaeological component, suggest that Monte Verde cannot be older than the Middle Holocene (8200 to 4200 years B.P.). With colonization no longer anchored by Monte Verde, our revised chronology supports a more recent date of human arrival to South America.

South America was the last continent colonized by humans, but the timing of that event is debated. The Monte Verde site in the Lake District of southern Chile (Fig. 1A) is central to this debate. The Monte Verde II (MV-II) component of Monte Verde was excavated from 1977 to 1985 (1, 2), yielding a small assemblage of lithic artifacts within a peat bed that preserved wood, fossil remains of extinct fauna, and other organic materials. The latter are interpreted to include wooden tools, architectural elements, quids of seaweed, and cordage. Previously dated to ~14,500 years before present (B.P.) (3), the identified occupation predates by nearly 1500 years securely dated North American Clovis contexts (4), which are long believed to mark the first peopling of the Americas (1, 2, 4–6). The previously identified age for MV-II also antedates by 500 years the oldest unambiguous sites in eastern Beringia (7–9), the presumed entry point to the Western Hemisphere (6, 8, 9).

In 1997, a team of archaeologists examined the Monte Verde collections, visited the site, and validated the MV-II component as the earliest securely dated archaeological evidence south of the continental ice sheets in the Americas (5, 10). Validated by external review, Monte Verde remains the discipline's most widely accepted pre-Clovis evidence, and its acceptance led to multiple changes in our understanding of the peopling of the Americas (6, 11–13). Further research has claimed to confirm the dating of MV-II, push back the age of human presence, and identify new archaeological components in the Monte Verde site complex (3, 14–16).

Although the archaeological community widely accepts the findings from Monte Verde, numerous researchers have questioned their validity. Some criticisms center on the associations between artifacts, strata, and dates, as well as reporting inconsistencies that complicate evaluation of the occupation's antiquity and cultural status (17, 18). Others have tested whether local reservoir effects or dating of old wood have caused dating inaccuracies at the site (3, 19). Researchers have noted that some paleoenvironmental records are more consistent with a Holocene than Pleistocene age (20, 21). Others have suggested the possibility that dated Pleistocene organic matter in the MV-II component was redeposited from older glacial outwash strata (21, 22). Last, some researchers have noted that the extreme range of radiocarbon dates reported for the site (23) is not consistent with a brief occupation as originally reported (20, 24, 25).

We reevaluated the age and formation of the MV-II archaeological component by describing, sampling, and dating nine alluvial exposures (Fig. 1B; figs. S5 to S15; and tables S1, S3, and S4). This study represents the first investigation of Monte Verde, independent of the original investigators, in the nearly 50 years since its discovery. Most of our sampling occurred along the banks of Chinchihuapi Creek, the drainage on which the site is located.

Because of differences in interpretation (supplementary text S2), we developed a stratigraphic scheme independent of the original investigators (26–29), who recognized two formations in the site area (supplementary text S5). They identified the older Salto Chico Formation as a regional glacial outwash deposit derived from the Late Pleistocene Llanquihue glaciation. They defined the younger Monte Verde Formation as a Late Pleistocene and Early Holocene alluvial terrace that preserves the MV-II component at its base. We refer to the sands and gravels of the Salto Chico Formation as stratigraphic unit 1a (SU1a). Conformably overlying SU1a are fine-grained clastic and organic sediments we call SU1b, -c, -d, and -e. SU1 strata are separated from Dillehay and Pino's (26–28) Monte Verde Formation (SU2 in our scheme) by an erosional contact (26, 28, 29). Unlike SU1, which occurs regionally, SU2 is an inset alluvial terrace (fig. S16) limited to Chinchihuapi Creek in the Monte Verde site area. The MV-II archaeological component occurs on bedload alluvium in lowermost SU2 (26, 28, 29).

Results

We identified four alluvial terrace surfaces (Fig. 1B). The lowest (T1) is the modern floodplain. T2 is composed mostly of SU2, with erosional remnants of SU1a at its base. The highest two surfaces in the valley, T3 and T4, are associated with SU1. In some locations, the upper surface of T3 is depositional; in others, it is erosional (fig. S17). On valley margins, T4 is always a depositional terrace and grades down to form the surface of T3 (Fig. 2A).

SU1 occurs across the area and forms the high surface in the vicinity of Chinchihuapi Creek (for example, the T4 and T3 surfaces) (Figs. 1B and 2A). Described as glacio-fluvial by Pino (26, 28), SU1a consists of well-rounded gravels and coarse to medium sand lenses, especially well exposed in sections 1, 3, and 5 (figs. S5, S7, S9). Although not described by Pino (26) as part of SU1 (supplementary text S2), the uppermost 60 to 90 cm of T4 (SU1b to SU1e) consist of fine clastic and organic sediments that conformably overlie SU1a (stained by organics at section 1) (fig. S5). There is no erosional contact between the lower and upper SU1 strata (supplementary text S2, S5, S9). SU1b is a wood layer that occurs along this contact (fig. S21). It is overlain by SU1c, a 10- to 20-cm-thick, very dark brown to black, organic clay (gyttja). Overlying is SU1d, a ~5-cm-thick ash, identified geochemically as the Lepué Tephra, a regional stratigraphic marker dated to 11,000 years B.P. (supplementary text S3, figs. S22 and S23, and data S1) (30–32). The tephra's age and composition eliminate the possibility of misidentification or mixing of multiple ashes (30, 31, 33). Although tephras have been noted at the site (3, 15, 26), the presence

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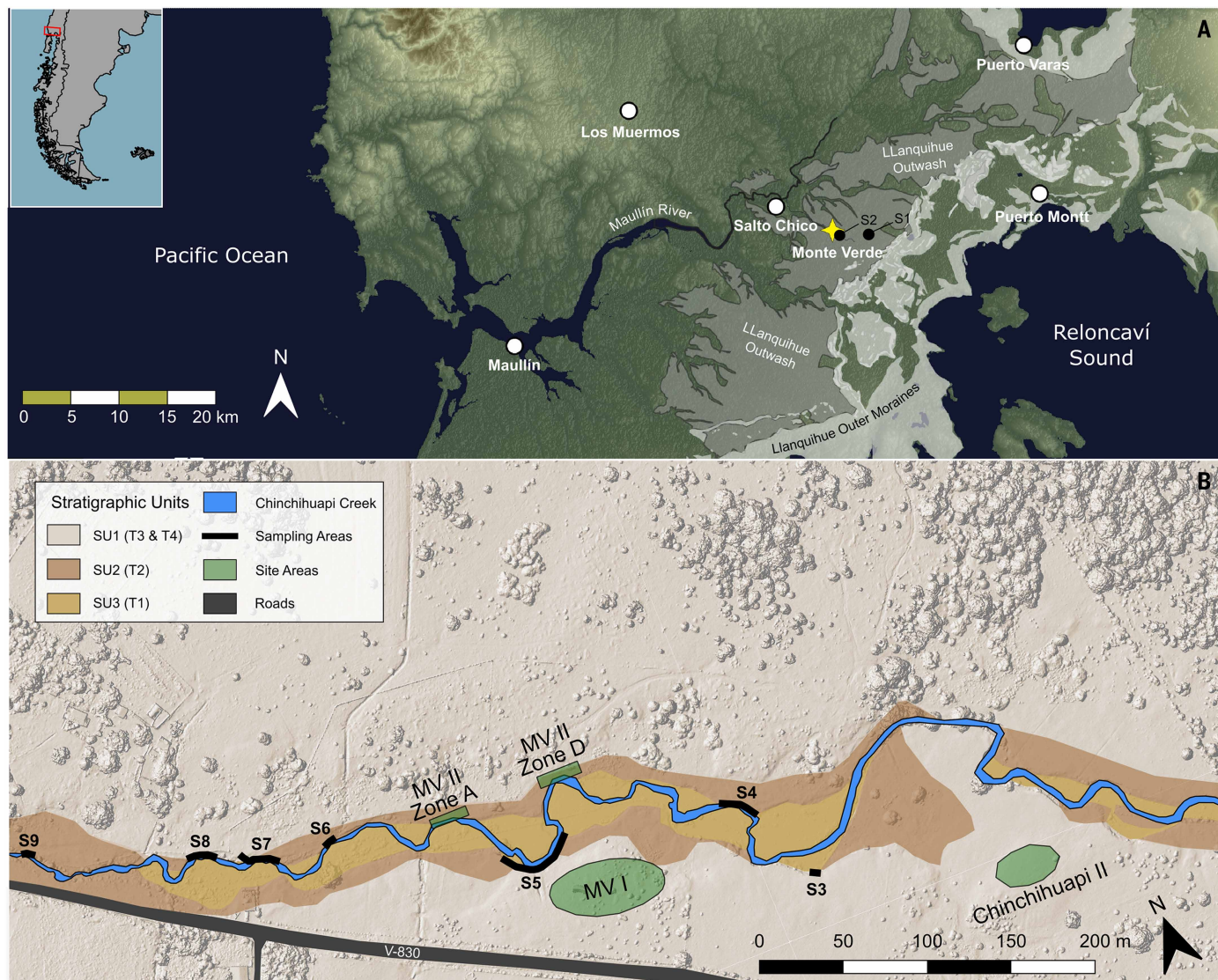


Fig. 1. Map of the study area. (A) Location of the Monte Verde site (yellow star) in the Lake District of southern Chile in relation to the terminal moraines from the last glaciation (light gray) and outwash plains (dark gray) grading to them. Locations of section1 (S1) and section2 (S2) are indicated with black dots. White dots indicate nearby towns. (B) Geologic map of the Monte Verde site area showing major geomorphic units and site areas from the original investigators in the Chinchihuapi valley. The base map is a digital elevation model created from drone photogrammetry.

of the Lepué Tephra, formally named in 2017, has not been previously recognized at Monte Verde. SU1d is overlain by the modern soil, formed in fine-grained sediments and peats (SU1e). Two dates on wood and bulk organic matter from this soil in section 1 are Late Holocene in age (Table 1).

Three radiocarbon dates from SU1a on wood from section 8 (Fig. 2 and fig. S13) produced ages between ~16,450 years (D-AMS 050109) and 16,170 years (D-AMS 050107). Wood and charcoal are abundant in SU1b (fig. S21), especially near the Monte Verde site; we dated this wood and charcoal in three different sections: one in the site area (section 5), one upstream along Chinchihuapi Creek (section 2), and one in a gravel quarry (section 1) a few kilometers away (figs. S5 and S9). They produced ages spanning 14,750 to 13,400 years, encompassing the previously identified age of the MV-II component (Fig. 2B). The overlying gyttja of SU1c is challenging to date because it contains abundant modern rootlets (fig. S21B), but radiocarbon ages consistently date to the Early Holocene from ~11,310 years (D-AMS-050100) to 9700 years B.P. (UCIAMS-292339). The 11,000-year-old Lepué Tephra

(SU1d) provides a terminus ante quem for the deposition of the underlying organic sediment (SU1c).

Direct ages on SU1a come from four single-grain OSL (optically stimulated luminescence) dates (Table 2 and table S5), two from the quarry (section 1), and one each from sections 4 and 9. These range from 23,000 years (VLL-0671-L & VLL-672-L) to 14,000 years B.P. (VLL-676-L). In total, SU1 accumulated until at least 11,000 years B.P., as indicated by the deposition of the Lepué Tephra, which stratigraphically underlies SU2 and thus provides a limiting age for the MV-II component (Fig. 2A).

The inset T2 surface is associated with SU2 deposition, although the bottom of the sedimentary units making up the terrace preserves locally SU1a sands and gravels (figs. S8, S10, S12, S14, and S16). The contact between these two strata at all sections is sharp and unambiguously erosional (26, 28, 29). SU2 is stratigraphically inset into SU1 (fig. S16). This disconformity represents missing time that began with initial incision of SU1, followed by later deposition of SU2. The two SUs are also easily distinguished by color. SU1a sands and gravels

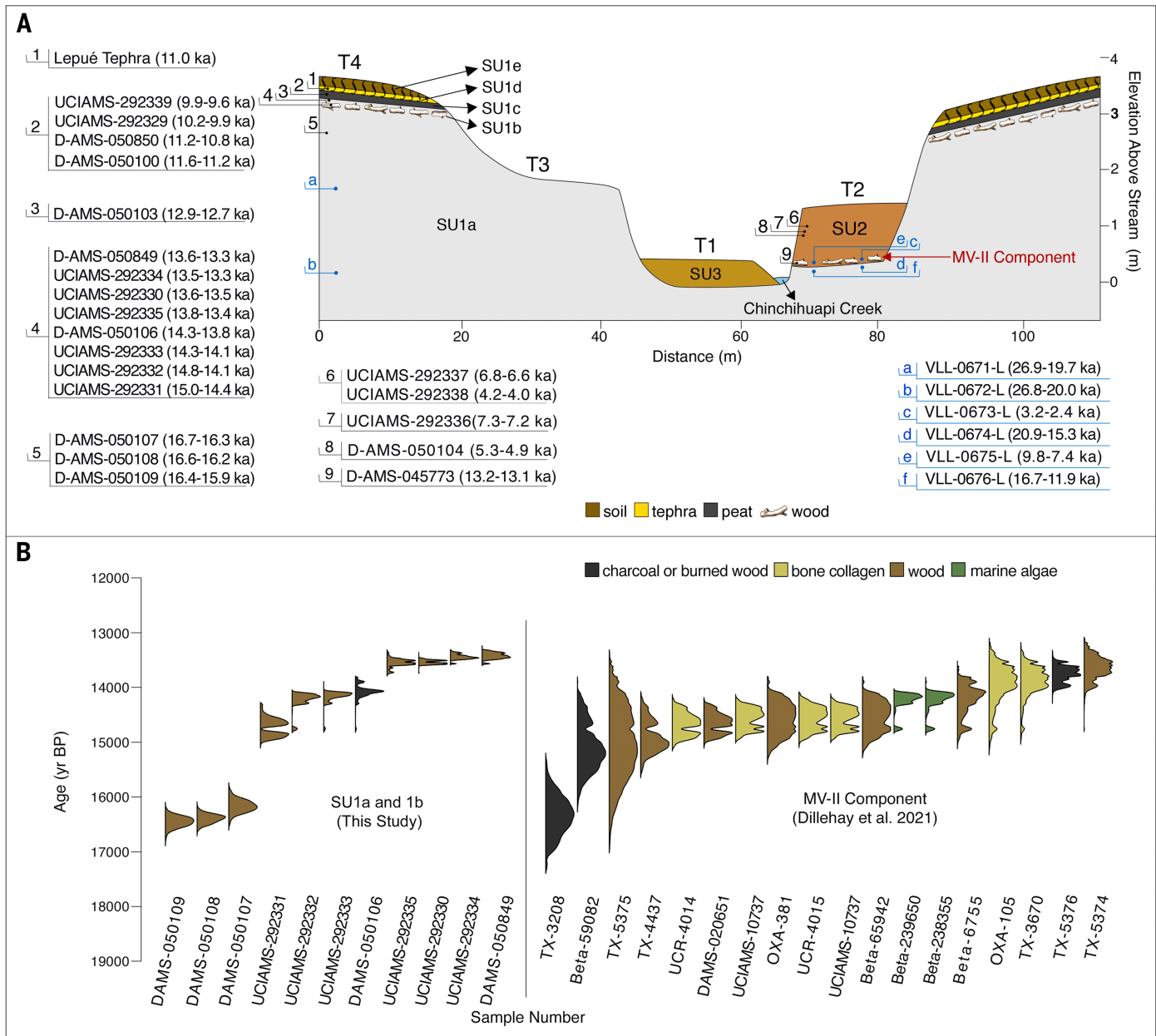


Fig. 2. Ages of geologic deposits and the Monte Verde site. (A) Generalized stratigraphic cross section of the Chinchihuapi valley showing stratigraphic position of the Lepué Tephra and all radiocarbon (black numbers) and OSL (blue letters) dates from this study. All dates are reported as 2σ calendar age ranges. **(B)** Comparison of radiocarbon dates on wood and charcoal from SU1a and SU1b from this study to those that are considered to best date the MV-II component at the base of SU2 in the Monte Verde site from Dillehay *et al.* (50).

usually exhibit gray to yellow hues (Munsell soil color notations 10YR 5/1 to 2.5YR 6/3) of glaciofluvial sediments, whereas those of SU2 are orange to brown (7.5YR 5/6 to 7.5YR 4/6), the result of iron oxide staining perhaps due to decomposition of anoxic peatlands (fig. S28) (34). Wood and peat “rip-up” clasts are abundant throughout SU2, indicating redeposition of older material into younger sediments. The MV-II cultural component rested on or just above the SU1/SU2 contact within SU2 (26, 28, 29).

Contrary to previous reports (26, 28, 29), we did not observe organic matter or peat near the SU1/SU2 contact, presumably because the portion of the site preserving the exact stratigraphy present in the MV-II excavations was lost to erosion (5). We were able to date a wood sample collected by C. V. Haynes on 9 January 1997 within SU2 slightly above the erosive contact (Table 1). That sample produced a Late

Pleistocene age of 13,150 years (D-AMS 045773), suggesting redeposition from SU1b into SU2. Radiocarbon dates on wood and peat from higher in SU2 within (section 6) and just outside of the site area (section 4) produced Middle Holocene ages between 7210 (UCIAMS-292336) and 4130 years (UCIAMS-292338) (Table 1). Two OSL ages directly date sands of SU2, one upstream (section 4) and one downstream (section 9) of the site (Table 2). These range between 8600 years (VLL-0675-L) and 2800 years B.P. (VLL-0673-L), confirming a Middle to Late Holocene age for this stratum and thus, we argue, the MV-II component.

Discussion

Our revised chronology and stratigraphy for the geological evolution of Chinchihuapi Creek (Fig. 3) agree well with regional glacial chronology and climate history. The Reloncaví glacial lobe retreated from the

Table 1. Radiocarbon dates from the Monte Verde site area.

Lab ID	Section	Stratum	Material	Depth (cm)	Age ± σ (¹⁴ C years B.P.)	Median calendar age [2σ range] (years B.P.)
D-AMS 050109	8	SU1a	wood	60	13,655 ± 57	16,450 [16,660 to 16,250]
D-AMS 050108	8	SU1a	wood	60	13,615 ± 42	16,390 [16,570 to 16,230]
D-AMS 050107	8	SU1a	wood	60	13,464 ± 73	16,170 [16,410 to 15,910]
UCIAMS-292331	5	SU1b	wood	55	12,525 ± 25	14,700 [15,000 to 14,430]
UCIAMS-292332	5	SU1b	wood	58	12,310 ± 25	14,190 [14,780 to 14,080]
UCIAMS-292333	5	SU1b	wood	42	12,275 ± 30	14,140 [14,320 to 14,050]
D-AMS 050106	1	SU1b	charcoal	40	12,207 ± 59	14,080 [14,310 to 13,810]
UCIAMS-292335	5	SU1b	wood	42	11,720 ± 60	13,540 [13,750 to 13,410]
UCIAMS-292330	5	SU1b	wood	46	11,715 ± 25	13,540 [13,600 to 13,470]
UCIAMS-292334	5	SU1b	wood	33	11,615 ± 25	13,450 [13,510 to 13,350]
D-AMS 050849	2	SU1b	wood	45	11,596 ± 48	13,420 [13,570 to 13,320]
D-AMS 045773	CVH	SU2	wood		11,275 ± 36	13,150 [13,240 to 13,100]
D-AMS 050103	3	SU1c	bulk organic matter	45	10,874 ± 47	12,770 [12,890 to 12,730]
D-AMS 050100	1	SU1c	gyttja	27	9,955 ± 47	11,310 [11,620 to 11,210]
D-AMS 050850	2	SU1c	gyttja	35	9,666 ± 37	10,950 [11,180 to 10,770]
UCIAMS-292329	5	SU1c	gyttja	28	8,920 ± 25	10,020 [10,180 to 9,890]
UCIAMS-292339	7	SU1c	gyttja	40	8,785 ± 25	9,720 [9,890 to 9,560]
UCIAMS-292336	6	SU2	brown peat	44	6,330 ± 25	7,210 [7,310 to 7,160]
UCIAMS-292337	6	SU2	wood	33	5,925 ± 20	6,710 [6,790 to 6,650]
D-AMS 050104	4	SU2	wood	50	4,506 ± 35	5,140 [5,300 to 4,890]
UCIAMS-292338	6	SU2	wood	34	3,805 ± 20	4,130 [4,240 to 3,990]
D-AMS 050099	1	SU1d	bulk organic matter	15	3,096 ± 27	3,270 [3,360 to 3,170]
D-AMS 050105	1	SU1d	wood	20	117 ± 29	

Table 2. Luminescence data for samples from the Monte Verde site area.

Sample lab code	Section	²³⁸ U (Bq/kg)	²³² Th (Bq/kg)	⁴⁰ K (Bq/kg)	Depth (cm)	KFs (n)*	σ _b (%)†	Fs IR50 De (Gy)‡	Dose rate (Gy/ka)§	IR50 age (ka) faded¶	g-value#	IR50 age (ka) fading corrected**
VLL-0671-L	1	10.78 ± 1.22	17.99 ± 0.21	265.02 ± 3.67	205	131	61 ± 4	36.26 ± 1.66	2.15 ± 0.13	16.9 ± 1.3	3.3 ± 0.03	23.3 ± 1.8
VLL-0672-L	1	12.39 ± 1.08	23.8 ± 0.26	245.72 ± 3.58	345	110	66 ± 5	35.29 ± 1.61	2.10 ± 0.12	16.8 ± 1.2	3.4 ± 0.02	23.4 ± 1.7
VLL-0673-L	4	7.94 ± 2.50	10.18 ± 0.17	160.51 ± 3.94	50	47	31 ± 4	3.62 ± 0.10	1.80 ± 0.13	2.0 ± 0.2	3.7 ± 0.04	2.8 ± 0.2
VLL-0674-L	4	9.26 ± 1.78	15.35 ± 0.2	286.34 ± 4.63	65	195	60 ± 4	26.61 ± 1.18	2.14 ± 0.14	12.4 ± 1.0	3.8 ± 0.02	18.1 ± 1.4
VLL-0675-L	9	11.44 ± 0.85	17.54 ± 0.19	265.61 ± 3.48	65	89	88 ± 9	14.27 ± 0.43	2.24 ± 0.13	6.4 ± 0.4	3.3 ± 0.03	8.6 ± 0.6
VLL-0676-L	9	11.31 ± 1.67	15.60 ± 0.19	300.50 ± 4.70	110	115	70 ± 5	21.05 ± 1.2	2.15 ± 0.33	9.8 ± 0.8	3.9 ± 0.04	14.3 ± 1.2

*Number of grains (aliquots for sample VLL-0673-L), passing the SAR rejection criteria. †Overdispersion calculated by using the CAM (51). ‡Calculated by using the bootstrapped MAM-3 (51,52). §Overall feldspar dose rate. External dose rate calculated using the conversion factors of Adamiec and Aitken (53) and the β-attenuation factors of Mejdahl (54), additionally taking an average water content of 20 ± 5% since burial and an average alpha efficiency (α-value) of 0.08 ± 0.02 into account. Internal dose rate calculated according to Huntley and Baril (55) by using an internal potassium content of 12.5 ± 0.5%. Cosmic radiation was calculated according to Prescott and Stephan (56) and Prescott and Hutton (57). ¶Age calculated with the software ADELE (58). #For each sample, 12 aliquots were analyzed for fading by using the approach of Auclair *et al.* (59); g-values are provided in percent per decade. **Corrected for fading according to the method of Huntley and Lamothé (60) by using the R Luminescence package (61).

outermost moraines by 25,000 years B.P. and collapsed into the Andes by 18,000 to 17,000 years B.P. (35). Pollen from nearby Lago Condorito shows cool and temperate conditions from 15,000 to 11,000 years B.P., when the former outwash surface was colonized by temperate rainforests (Fig. 3A) (32). This forest gave way to bogs as gyttja and peat infilled. Those organic sediments were buried by the 11,000-year-old Lepué Tephra (Fig. 3B). Incision of Chinchihuapi Creek began after, likely coeval with an extreme dry and warm phase 11,000 to 7600 years B.P. (Fig. 3C) (32). As glaciers retreated, isostatic adjustment would have further promoted incision by lowering base level (36). Deposition

of SU2 began at the onset of the Middle Holocene (Fig. 3D) after substantial incision of SU1 (2.5 to 3 m), including the uppermost 1 m of fine sediments of in situ Late Pleistocene strata containing abundant wood and peat (SU1b and SU1c). These results fail to support the hypothesis that the lower portion of SU2 and MV-II component date to the Late Pleistocene. Instead, evidence from multiple sections show that the uppermost terrace (SU1) at Monte Verde accumulated during the Late Pleistocene and Early Holocene. Its upper meter (SU1b and SU1c) contains abundant wood and organic matter capped by the Lepué Tephra (SU1d)

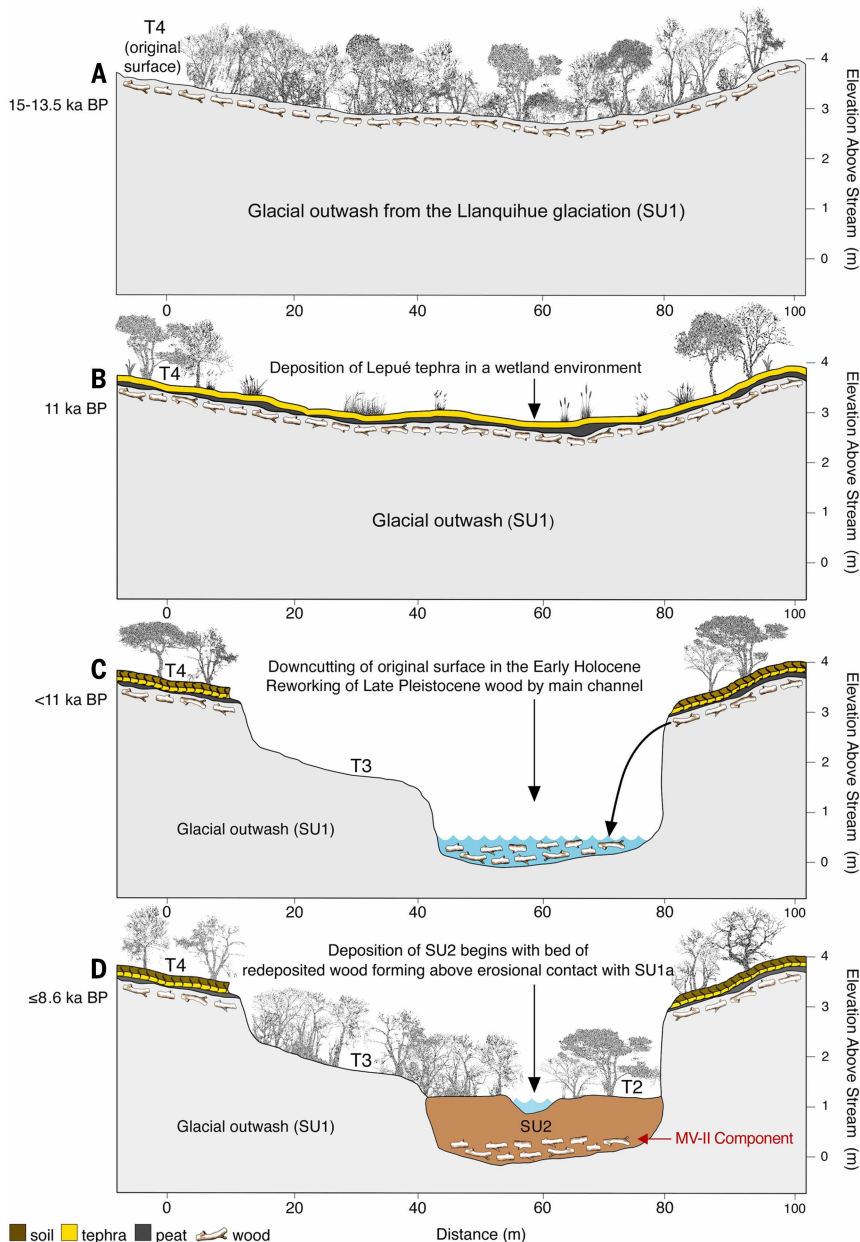


Fig. 3. Late Quaternary evolution of the Chinchihuapi valley. (A) At 15,000 to 14,000 years B.P., temperate rainforest filled a shallow swale on the former glacial outwash plain (SU1a and SU1b) dating from ~26,000 to 15,500 years B.P. in the valley. (B) A bed of wood from the forest that previously occurred in the valley center is buried by organic marsh sediments and later by Lepuë Tephra at 11,000 years B.P. (C) In the Early and Middle Holocene, after episodes of alluvial incision, wood and peat are redeposited onto the eroded surface of SU1 in the channel and bars of Chinchihuapi Creek. (D) SU2 aggraded, burying the “old” wood, peat, and clastic sediment in the MV-II component [compare with figure 5.35 from Pino (26)].

at 11,000 years B.P. (30). This tephra is a well-dated regional chronostratigraphic marker that is older than the MV-II component; additionally, it is absent in SU2, as would be expected in an aggrading system spanning the Pleistocene-Holocene transition. When alluvial deposition resumed after ~8600 years B.P., Pleistocene wood, megafaunal remains, and other organic matter present in SU1 in the immediate site area were reworked by Chinchihuapi Creek into Middle Holocene alluvium. As the stream undercut its banks, wood from SU1a and SU1b accumulated in the channel and on point bars. Thus, we conclude that the surface on which the MV-II component rests

did not exist in the Late Pleistocene and is a product of Early Holocene erosion. Although the organic-rich deposits of the MV-II component can produce Late Pleistocene radiocarbon ages, this material was redeposited after 8600 years B.P.

Previously published radiocarbon ages for SU1 and SU2 from Monte Verde are consistent with this hypothesis (supplementary text S9) because they suggest 7300 years of overlap (table S7 and fig. S35), which we argue can only be explained by redeposition of dated materials. Redeposition continues today in Chinchihuapi Creek, which regularly transports wood in its channel (supplementary text S6 and figs. S29 to S31). Our site-formation model supports previous researchers’ arguments (20, 24, 25) that the MV-II radiocarbon dates (Fig. 2B) are inconsistent with a brief occupation but instead represent an accumulation of redeposited materials (supplementary text S7).

Our revised age estimate for the MV-II component is also chronologically and typologically consistent with regional archaeology (supplementary text S8 and fig. S31). Although regional archaeological radiocarbon records show that Holocene date distributions closely match high-resolution fire records (37), Late Pleistocene dates do not. During Monte Verde’s alleged Pleistocene occupation, charcoal records show a substantial lull in fire activity (fig. S33), suggesting that humans were absent. Furthermore, MV-II diagnostic lanceolate bifacial points are very similar to materials that appear by 6400 years B.P. in regional archaeological sites (fig. S34) (38). The same is true of polished slate artifacts, which are ubiquitous in shell midden sites of Middle to Late Holocene age along the coast of the Reloncaví Sound (39). Other lithic materials such as thick large bifaces, notches, and choppers are sufficiently informal to be common at Holocene hunter-gatherer sites in the region (40).

Conclusions

The original interpretation of an age of 14,500 years for the Monte Verde MV-II component is challenged by three observations we bring to light: (i) The 11,000-year-old Lepuë Tephra (SU1d) is stratigraphically below and older than the MV-II component. (ii) There is an erosional contact representing missing time that separates SU1 and SU2, but there is no missing time in the site-formation scenario of the original investigators (15). (iii) Abundant Pleistocene wood and organic matter in the site area in the top of T4 (SU1a and SU1b) is available for redeposition that is the same age as wood in the MV-II component (Fig. 2B). Our interpretation of site formation accommodates all three observations. After deposition of

the Lepuë Tephra and downcutting of Chinchihuapi Creek during the Early Holocene, Pleistocene organic matter was redeposited and buried in alluvial sediments dated by means of OSL to the Middle Holocene, forming the MV-II component.

We conclude that the MV-II component of the Monte Verde site is Middle Holocene in age, and therefore, any inferences made about the peopling of the Americas based on a Late Pleistocene date for this site should be reevaluated. Although a pre-Clovis human presence in the Americas is accepted by many archaeologists (5, 6, 12), a Holocene age for Monte Verde leaves open the possibility of later initial colonization.

The acceptance of the pre-Clovis age of Monte Verde led some to reject migration through the ice-free corridor as a possible route of initial entry, and a coastal route along the Cordilleran ice margin has been suggested to be more likely (11, 13, 41). Although our findings do not preclude the possibility of earlier dates of initial entry to the Americas, they do support an initial interior migration into continental North America as a viable colonization hypothesis (42). As demonstrated here, the age of the MV-II component should not be used as a constraint or check on colonization models derived from other sources, including the genetics of modern or ancient populations (43–46). Our findings also underscore the critical need for independent study and verification of early sites (47–49).

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S35; Tables S1 to S7; References (62–144); Data S1; Reproducibility Checklist

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A mid-Holocene age for Monte Verde challenges the timeline of human colonization of South America

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Editor's summary

There is much debate surrounding the dates of human occupation in the Americas. One site that places South American timing in the pre-Clovis period is Monte Verde II in Chile. This site, however, was mostly only studied by a single group, meaning that replication of the dating has been lacking. Surovell *et al.* independently examined the site using current approaches and a broader scope and concluded that the site could not be older than 8200 years, significantly younger than the previously determined 14,500 years (see the Perspective by Rech). —Sacha Vignieri

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